

Question 9: Write a Python program to generate 1000 random numbers from a normal distribution with mean = 50 and standard deviation = 5. Compute its mean and standard deviation using NumPy, and draw a histogram to visualize the distribution.

```
import numpy as np
import matplotlib.pyplot as plt

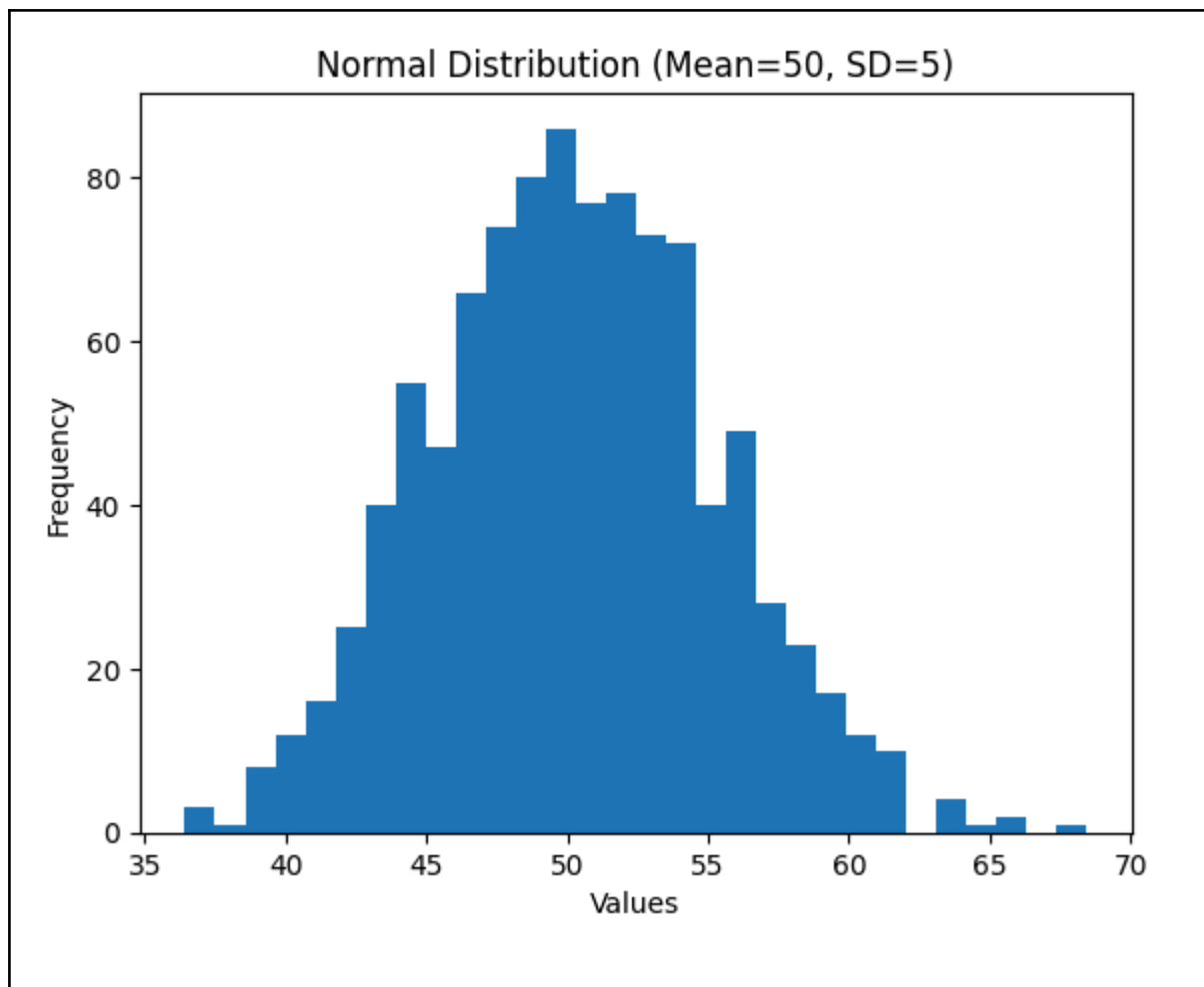
data = np.random.normal(50, 5, 1000)

# Compute mean and standard deviation
mean = np.mean(data)
std_dev = np.std(data)

print("Mean:", mean)
print("Standard Deviation:", std_dev)

plt.hist(data, bins=30)
plt.title("Normal Distribution (Mean=50, SD=5)")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()

Mean: 50.256511090266784
Standard Deviation: 5.002162168127246
```



You are working as a data analyst for a retail company. The company has collected daily sales data for 2 years and wants you to identify the overall sales trend.

```
daily_sales = [220, 245, 210, 265, 230, 250, 260, 275, 240, 255,  
               235, 260, 245, 250, 225, 270, 265, 255, 250, 260]
```

- Explain how you would apply the Central Limit Theorem to estimate the average sales with a 95% confidence interval.
- Write the Python code to compute the mean sales and its confidence interval.

```
import numpy as np
import scipy.stats as stats

daily_sales = [220, 245, 210, 265, 230, 250, 260, 275, 240, 255,
               235, 260, 245, 250, 225, 270, 265, 255, 250, 260]

data = np.array(daily_sales)

mean = np.mean(data)
std_err = stats.sem(data)

confidence = 0.95
interval = stats.t.interval(confidence, len(data)-1, loc=mean, scale=std_err)

print("Mean Sales:", mean)
print("95% Confidence Interval:", interval)

Mean Sales: 248.25
95% Confidence Interval: (np.float64(240.1695702514714),
np.float64(256.3304297485286))
```